

HW 2

1

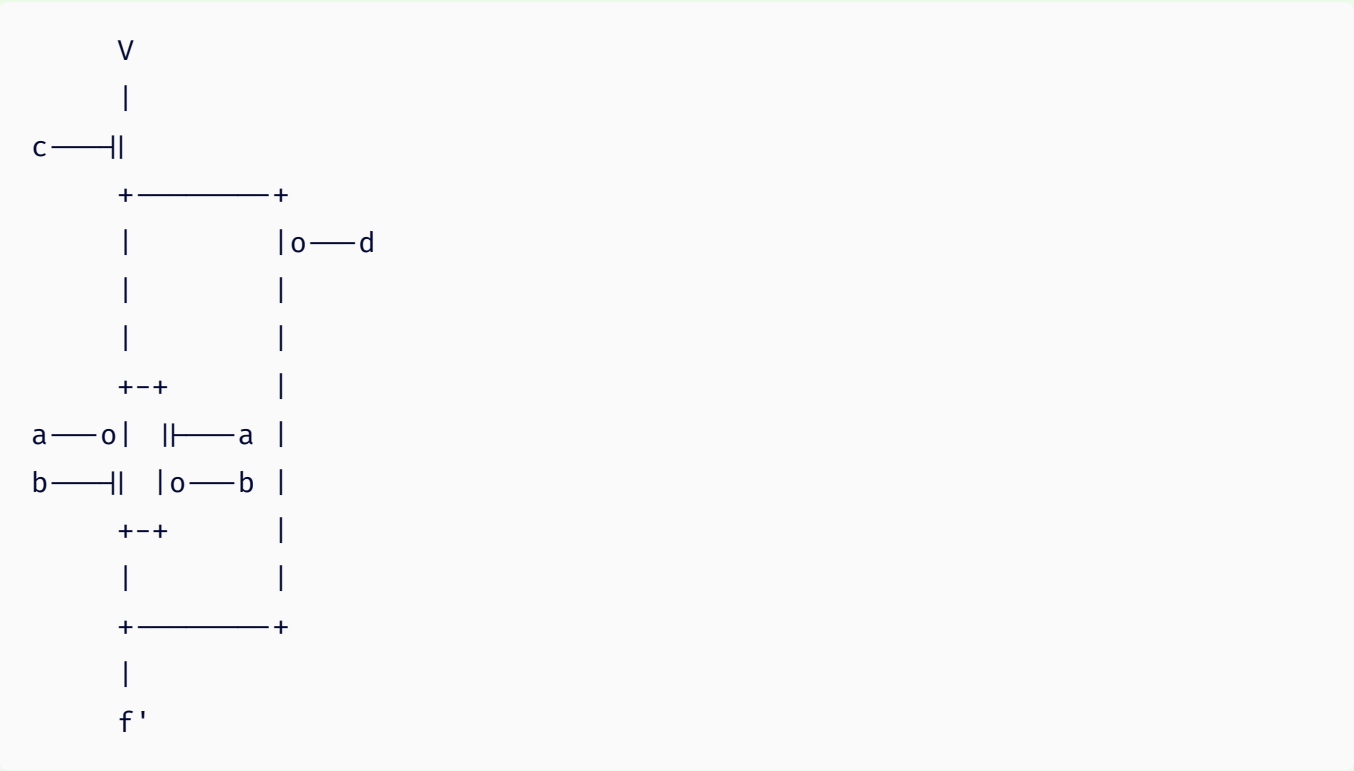
Construct the pass network connection diagram of a freeform tree network realization of the Boolean function given below. The tree network will connect to the input of an inverter. The inverter input will be f . The inverter output will f' . Use as few devices in the tree network as possible.

$$f(a,b,c,d) = \sum(0, 5, 7, 9, 11, 12) + d(1, 2, 4, 8, 13, 14)$$

✓ Answer ✓

cd\ab	00	01	11	10
00	1	X	1	X
01	X	1	X	1
11	0	1	0	1
10	X	0	X	0

$$f = c' + d \cdot (a \odot b)'$$



2

Given:

$$z = \overline{g1((ABC + ABD + ACD + BCD)) + g2}$$

$$\overline{g1} = ABCD$$

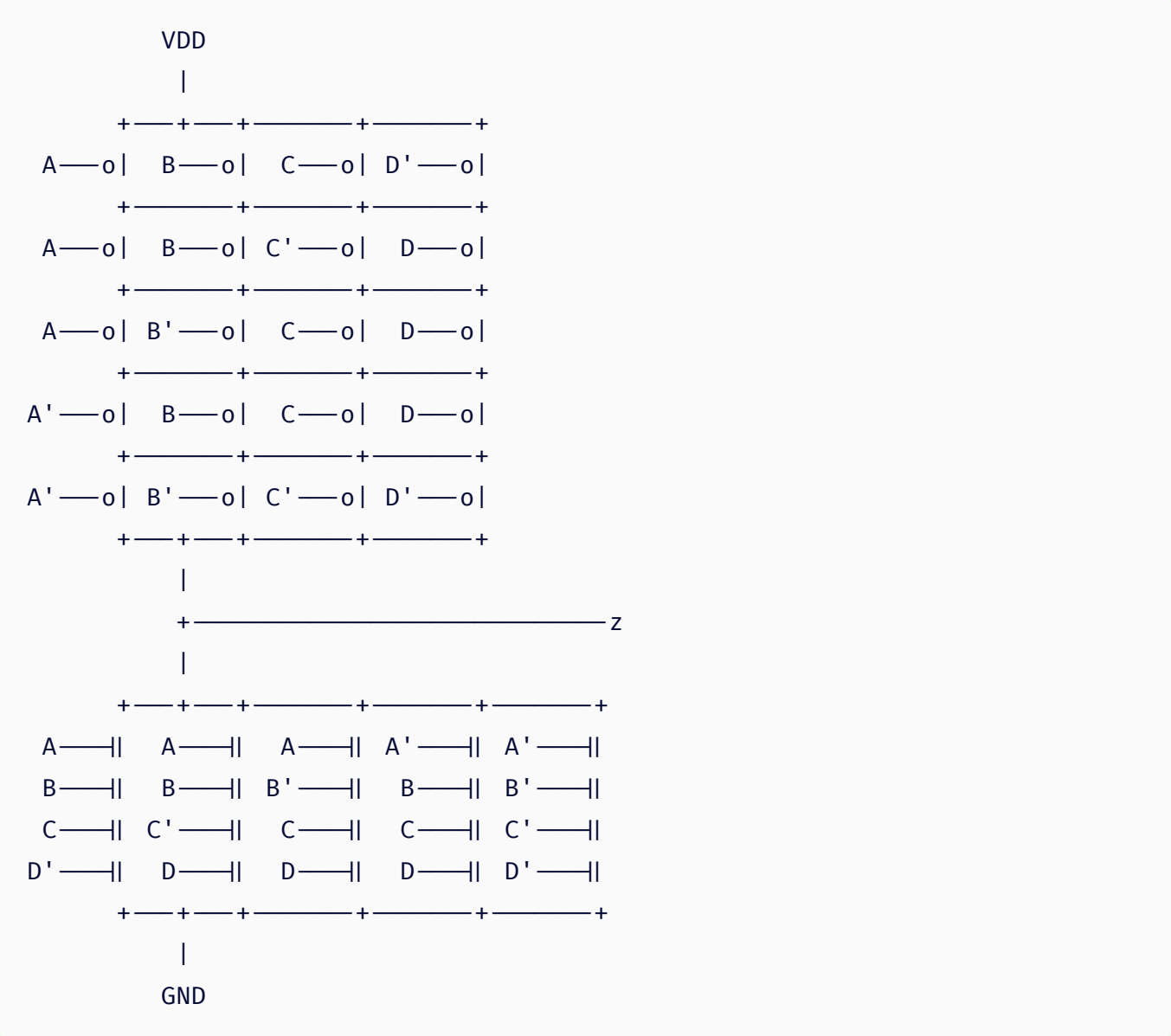
$$\overline{g2} = A + B + C + D$$

Obtain a switch diagram realization of the negative gate expression in

a

Complex Static CMOS gate realization

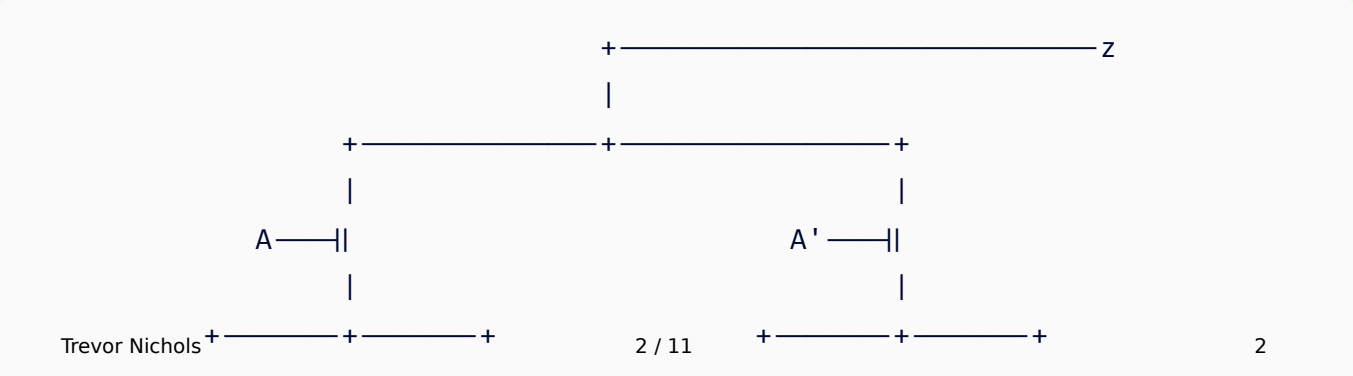
✓ Answer

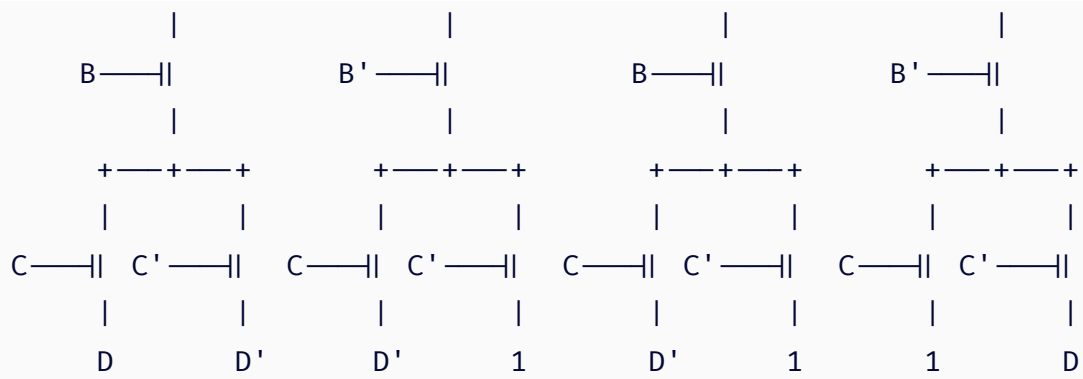


b

Pass transistor realization using A, B, C as control signals and '0', '1', D as data signals.

✓ Answer

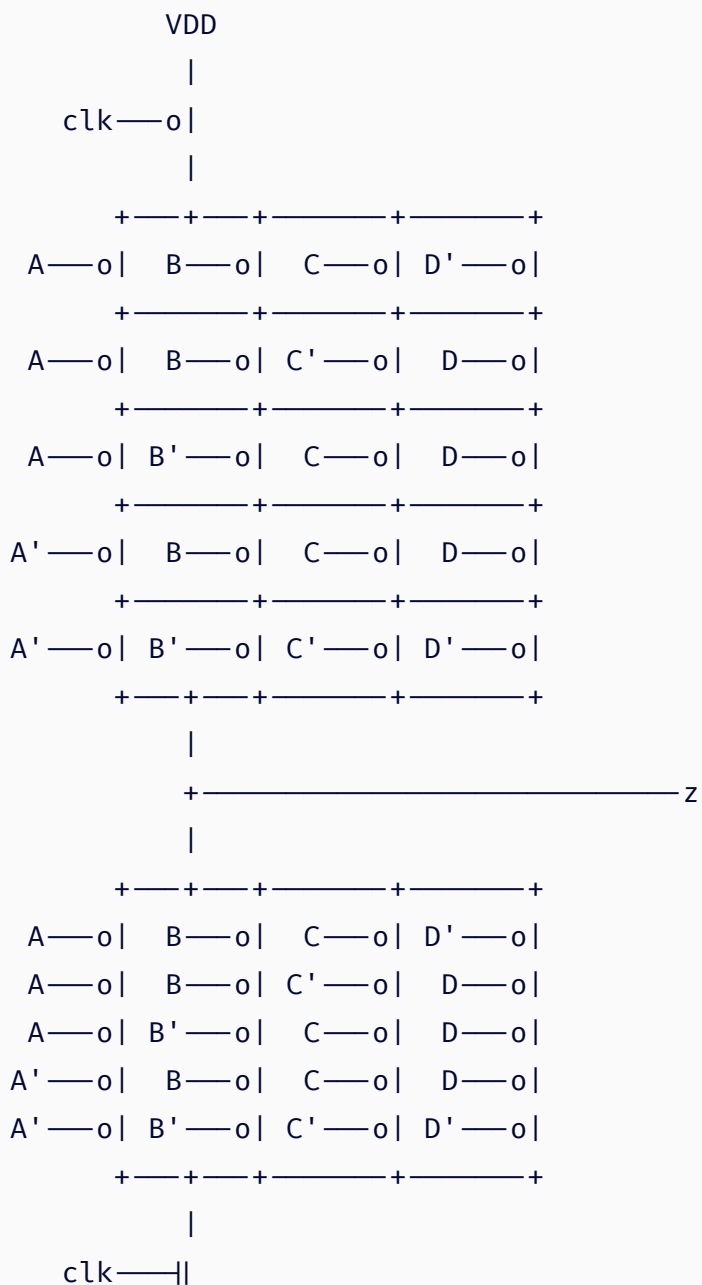




C

Dynamic CMOS realization

✓ Answer



|
GND

3

The simple NMOS pass network into a CMOS inverter given below might seem to pass for a suitable CMOS realization of the function $AC + \overline{AB}$. Does this circuit satisfy the CMOS assertion that no power is consumed in steady state? Why or why not? Will the output voltage be degraded from nominal values?

✓ Answer

There is no leakage during steady state as there is no path from VDD to ground when fully switched.

There may be voltage degradation as the NMOS circuit before the CMOS inverter may not have enough voltage to close the PMOS VDD gate at the end, however, due to NMOS voltage degradation.

4

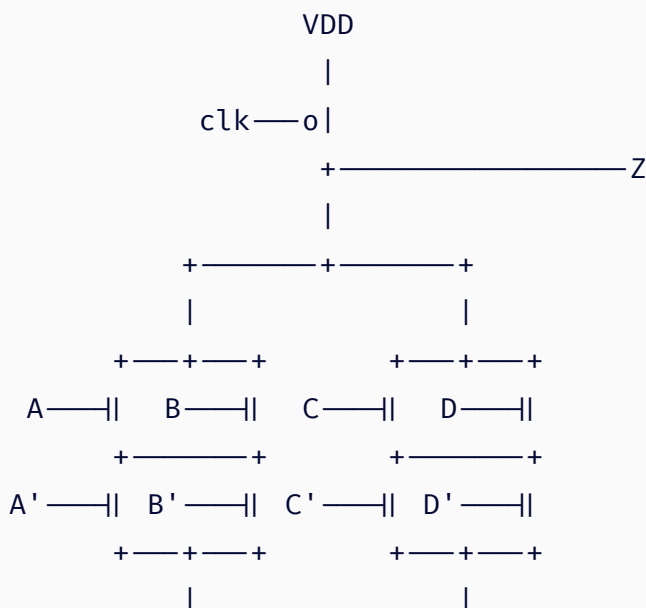
Consider the function $G = ABCD + ABC'D' + A'B'CD + A'B'C'D'$ which functions as a comparator: when $A = B$ and $C = D$ it produces a logic 1. This can also be written as $(AB + A'B')(CD + C'D')$. Draw the transistor-level schematic for G using:

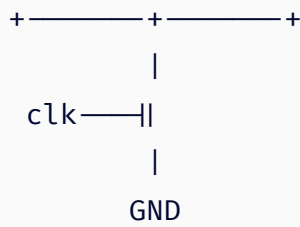
a

N-Domino

✓ Answer

$$G' = ((A + B)(A' + B')) + ((C + D)(C' + D'))$$



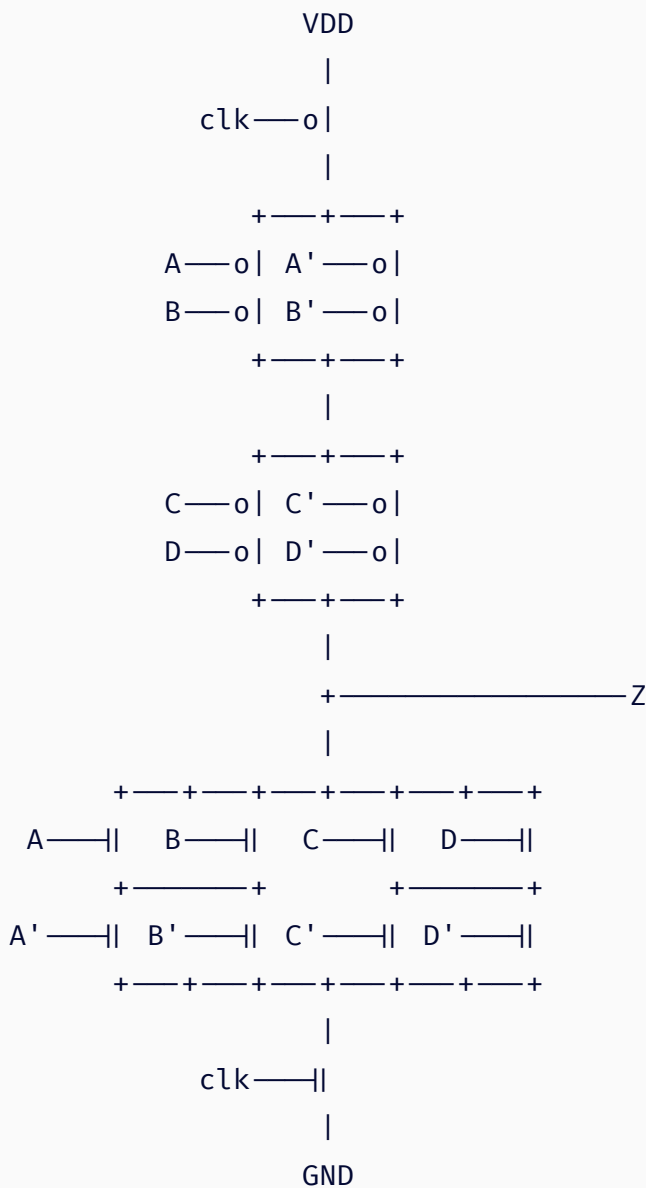


b

NP Domino

✓ **Answer**

$$G' = (AB + A'B')(CD + C'D')$$



5

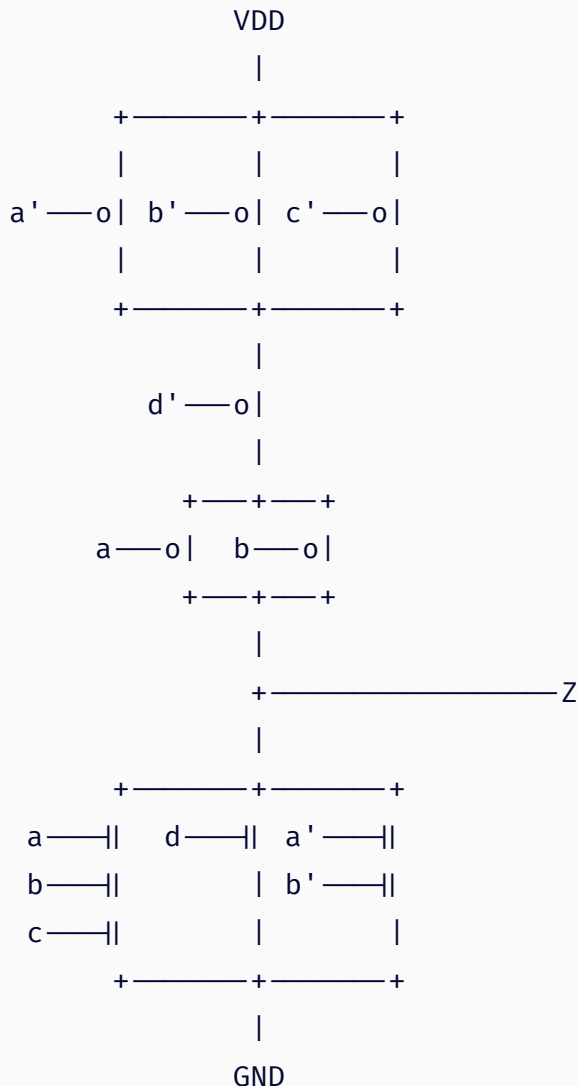
Implement $F = \overline{(a + c + b)dab}$ using CMOS logic with the minimum number of transistors.

✓ Answer

$$F = (a + c + b)\overline{dab}$$

$$F = ab + d' + a'b'c'$$

$$F' = (a + b + c)d(a' + b')$$

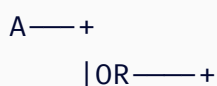


6

a

Convert this design into one that lends itself to a domino CMOS implementation. Do not write out logic equations, minimize gates or merge signals to create complex gates without rewiring, but merely by changing the existing gates and adding and removing inverters, you are to create 5 domino gates that compute the same value for Z as the circuit above. Use Demorgan's Law to transform existing gates where necessary. The inputs (A , B , D and E) are driven by domino gates and the output, Z , must be able to drive a domino gate.

✓ Answer



```

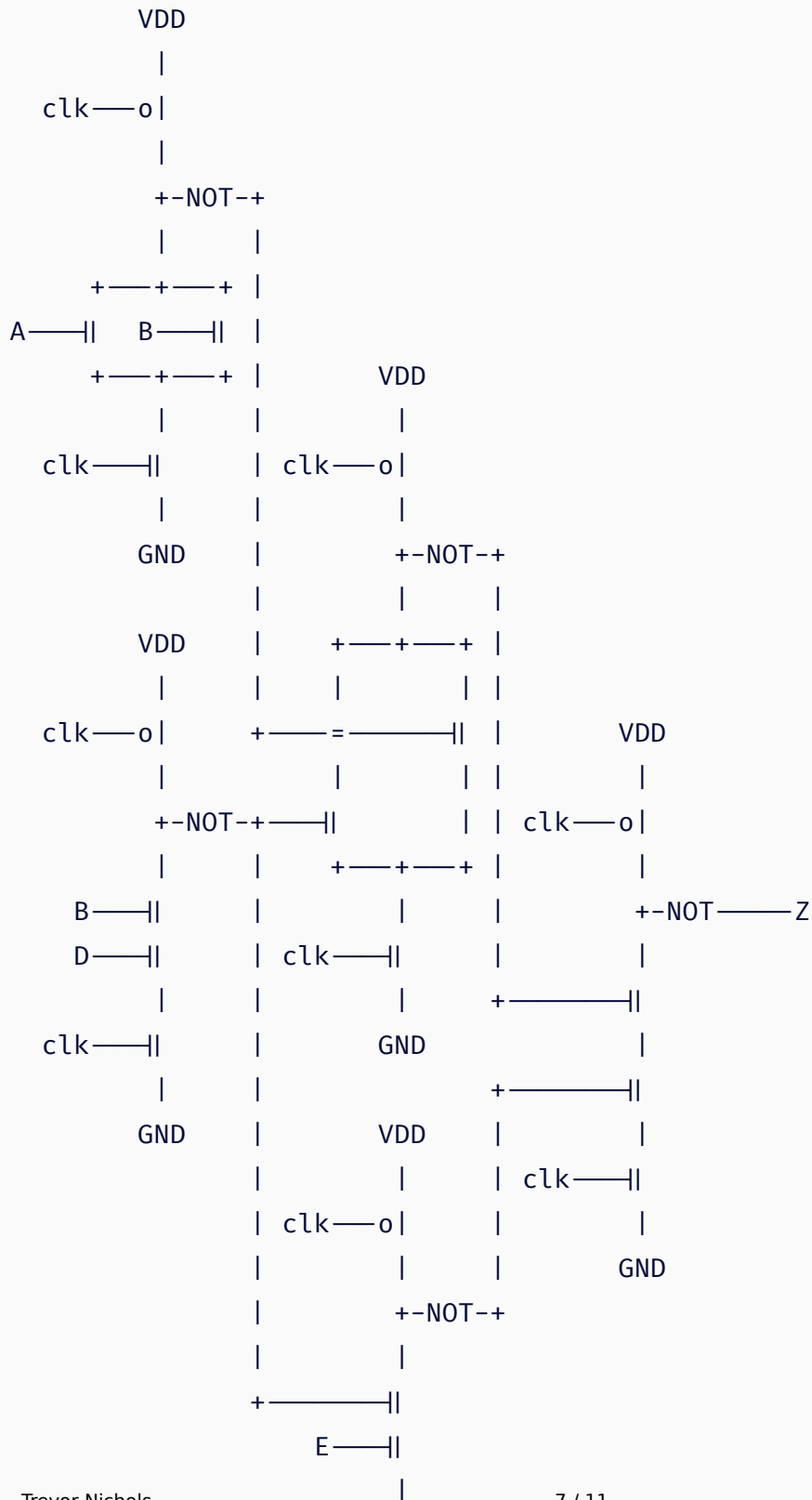
B —+      | OR —+
      | AND —+      +AND — Z
D —+      | AND —+
E —————+

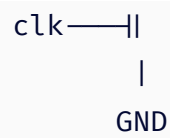
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b

Draw a transistor-level schematic for your entire domino circuit.

✓ Answer

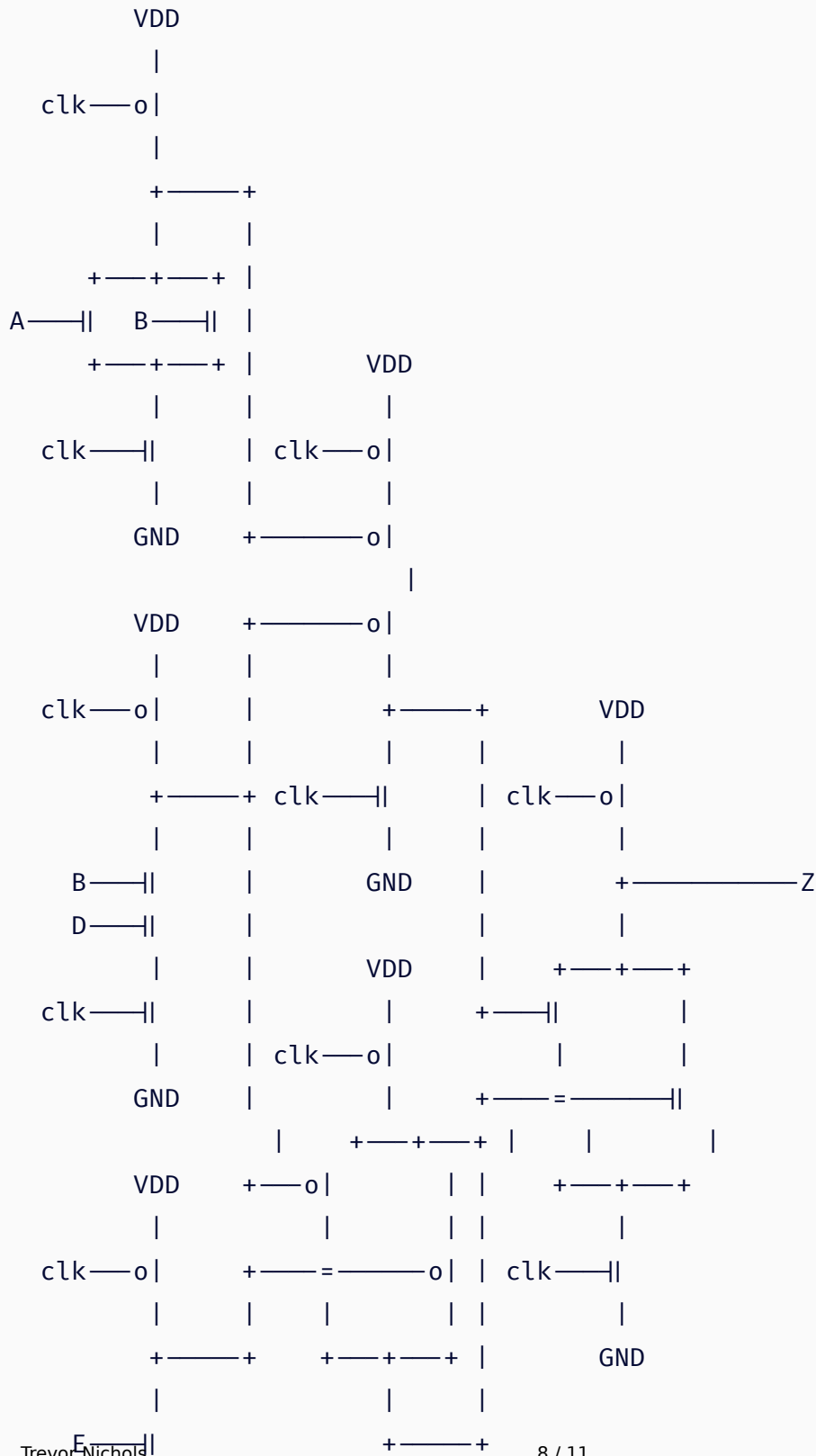


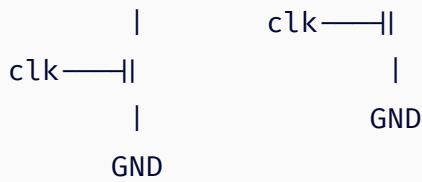


C

Using the part 'A)' approach, draw a transistor-level NP domino circuit for the gate-level circuit shown. Assume that the stages that generated the inputs (*A*, *B*, *D*, and *E*) are pre-charged low.

✓ Answer





7

Minimize the following Boolean equations to eliminate redundancy

a

$$ab + bcd + c\bar{a}$$

✓ Answer

$$b(a + cd) + c\bar{a}$$

b

$$(x + y)(x + z)$$

✓ Answer

$$x + yz$$

c

$$a(b + c + d) + b(c + d + a) + c(d + a + b) + d(a + b + c)$$

✓ Answer

$$a(b + c + d) + b(c + d) + cd$$

8

Label the inputs of the static CMOS circuit shown so that it implements

$$Z = \overline{ac + a\bar{c}bd + f + aef}$$

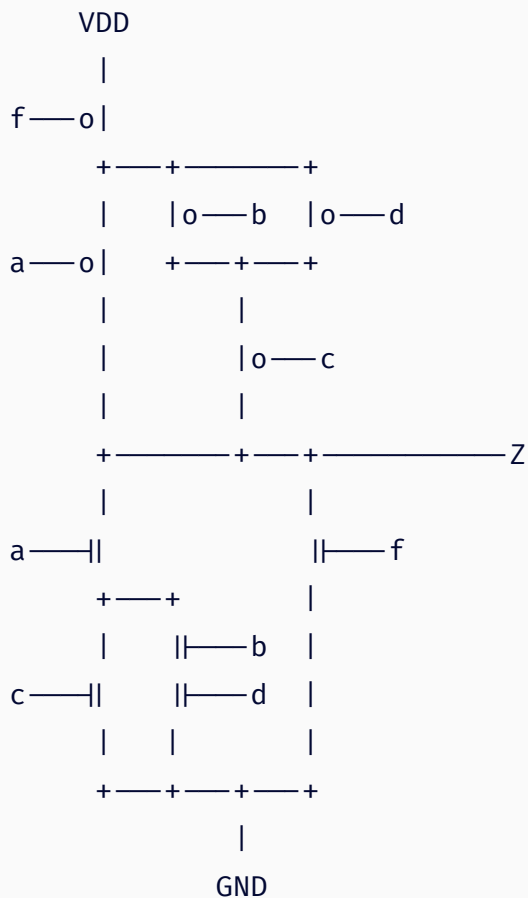
✓ Answer

$$Z = \overline{ac + a\bar{c}bd + f + aef}$$

$$Z = \overline{ac + abd + f}$$

$$Z = (a' + c')(a' + b' + d')f'$$

$$Z = f'(a' + c'(b' + d'))$$



9

Draw a complete transistor level diagram to implement the function below. Use static CMOS logic. Assume that the signals and their complements are available. Use compound gates where possible. You do not need to minimize the logic. Please connect the output wires from one gate to the inputs of the next gate, all in the same diagram. Exclusive-OR is shown as $\otimes$.

$$Z1 = \overline{ab + c + de}$$

$$Z2 = \overline{f \otimes g}$$

$$Out = \overline{Z1Z2}$$

✓ Answer

$$Z = \overline{ab + c + de} \overline{f \otimes g}$$

$$Z = (ab + c + de) + (f \otimes g)$$

$$Z' = ((a' + b')c'(d' + e'))(f' \otimes g)$$

